

Worksheet — Polynomials (Class 10 CBSE Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show all working. Total time: 50 minutes.

Q1. The degree of $5x^3 - 7x^2 + 2x - 9$ is: [1 mark]

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q2. A polynomial of degree n has at most n real zeros. [1 mark]

- (A) True
- (B) False

Q3. If α and β are zeros of $ax^2 + bx + c$, then $\alpha\beta = \underline{\hspace{2cm}}$. [1 mark]

Q4. For $x^2 - 5x + 6$, the sum of zeros is: [1 mark]

- (A) -5
- (B) 5
- (C) 6
- (D) -6

Q5. If a quadratic has discriminant 0 , its graph: [2 marks]

- (A) crosses the x -axis at two points
- (B) touches the x -axis at one point
- (C) doesn't cross the x -axis
- (D) is a straight line

Q6. Find the zeros of $x^2 - 5x + 6$ by factorisation, and verify the sum-and-product relations. [3 marks]

Q7. If α and β are zeros of $x^2 - 7x + 10$, find: (i) $\alpha + \beta$, (ii) $\alpha\beta$, (iii) $\alpha^2 + \beta^2$, (iv) $1/\alpha + 1/\beta$. [3 marks]

Q8. Construct a quadratic whose zeros are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. [3 marks]

Q9. If α, β, γ are zeros of $x^3 + 2x^2 - 5x - 6$, find: (i) $\alpha + \beta + \gamma$, (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$, (iii) $\alpha\beta\gamma$. [3 marks]

Q10. Divide $p(x) = 3x^3 + x^2 + 2x + 5$ by $g(x) = x^2 + 2x + 1$ using the division algorithm. Show every step. Verify $p = g \cdot q + r$. [6 marks]

Q11. Two zeros of $x^3 - 4x^2 + x + 6$ are 2 and -1 . Find the third zero using the division algorithm. [6 marks]

Answer Key

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Q1. The degree of $5x^3 - 7x^2 + 2x - 9$ is:

Answer: 3

Highest power.

MCQ · EASY · 10-CBSE-MATH-PO-01

Q2. A polynomial of degree n has at most n real zeros.

Answer: True

Standard theorem.

TRUE_FALSE · EASY · 10-CBSE-MATH-PO-02

Q3. If α and β are zeros of $ax^2 + bx + c$, then $\alpha\beta = \underline{\hspace{2cm}}$.

Answer: c/a

Vieta.

FILL_BLANK · EASY · 10-CBSE-MATH-PO-04

Q4. For $x^2 - 5x + 6$, the sum of zeros is:

Answer: 5

$-b/a$.

MCQ · EASY · 10-CBSE-MATH-PO-04

Q5. If a quadratic has discriminant 0, its graph:

Answer: touches the x-axis at one point

Equal real zeros.

MCQ · MEDIUM · 10-CBSE-MATH-PO-03

Q6. Find the zeros of $x^2 - 5x + 6$ by factorisation, and verify the sum-and-product relations.

Answer: $x^2 - 5x + 6 = (x - 2)(x - 3)$; zeros are 2 and 3. Sum = 5 = $-b/a$ ✓. Product = 6 = c/a ✓.

Factorise + verify.

SHORT_ANSWER · MEDIUM · 10-CBSE-MATH-PO-04

Q7. If α and β are zeros of $x^2 - 7x + 10$, find: (i) $\alpha + \beta$, (ii) $\alpha\beta$, (iii) $\alpha^2 + \beta^2$, (iv) $1/\alpha + 1/\beta$.

Answer: (i) 7. (ii) 10. (iii) $(7)^2 - 2(10) = 49 - 20 = 29$. (iv) $(\alpha+\beta)/(\alpha\beta) = 7/10$.

Symmetric functions of zeros.

SHORT_ANSWER · MEDIUM · 10-CBSE-MATH-PO-04

Q8. Construct a quadratic whose zeros are $3 + \sqrt{2}$ and $3 - \sqrt{2}$.

Answer: Sum = 6; product = 9 – 2 = 7. Quadratic: $x^2 - 6x + 7$.

Use sum/product $\rightarrow$ standard form.

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Q9. If α, β, γ are zeros of $x^3 + 2x^2 - 5x - 6$, find: (i) $\alpha + \beta + \gamma$, (ii) $\alpha\beta + \beta\gamma + \gamma\alpha$, (iii) $\alpha\beta\gamma$.

Answer: a = 1, b = 2, c = -5, d = -6. (i) $-b/a = -2$. (ii) $c/a = -5$. (iii) $-d/a = 6$.

Cubic Vieta.

SHORT_ANSWER · MEDIUM · 10-CBSE-MATH-PO-06

Q10. Divide $p(x) = 3x^3 + x^2 + 2x + 5$ by $g(x) = x^2 + 2x + 1$ using the division algorithm. Show every step. Verify $p = g \cdot q + r$.

Answer: Step 1: $3x^3 \div x^2 = 3x$. $3x \cdot (x^2 + 2x + 1) = 3x^3 + 6x^2 + 3x$. Subtract from p: $(3x^3 + x^2 + 2x + 5) - (3x^3 + 6x^2 + 3x) = -5x^2 - x + 5$. Step 2: $-5x^2 \div x^2 = -5$. $-5 \cdot (x^2 + 2x + 1) = -5x^2 - 10x - 5$. Subtract: $9x + 10$ (degree 1 < 2; STOP). Quotient = $3x - 5$; remainder = $9x + 10$. Verify: $g \cdot q + r = (x^2 + 2x + 1)(3x - 5) + (9x + 10) = 3x^3 + x^2 + 2x + 5 \checkmark$.

Standard worked example.

LONG_ANSWER · HARD · 10-CBSE-MATH-PO-07

Q11. Two zeros of $x^3 - 4x^2 + x + 6$ are 2 and -1. Find the third zero using the division algorithm.

Answer: Divide $x^3 - 4x^2 + x + 6$ by $(x - 2)(x + 1) = x^2 - x - 2$. Long division: $x^3 \div x^2 = x$. $x \cdot (x^2 - x - 2) = x^3 - x^2 - 2x$. Subtract: $-3x^2 + 3x + 6$. $-3x^2 \div x^2 = -3$. $-3 \cdot (x^2 - x - 2) = -3x^2 + 3x + 6$. Subtract: 0. So quotient = $x - 3$; third factor $(x - 3) \rightarrow$ third zero is 3.

Use known zeros + division to find missing zero.

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